Project Description

Project Title: Pharmaceutical Inventory Optimization & Financial Risk Analysis

Background & Context:

In the pharmaceutical distribution industry, inventory management is heavily constrained by strict product expirations, specific cold chain storage requirements, and high holding costs. Poor visibility into stock health often leads to a 'Profitability Paradox'—where high sales figures are completely neutralized by massive overhead and operational waste.

The Core Challenge:

This project addresses a severe financial deficit where a warehouse generates **$185.94M in Total Sales** with a **25.88% Gross Margin ($48.11M)**, yet results in a devastating **Net Profit of -$27.16M**. The root cause analysis identifies two major leaks:

1. **Excessive Storage Expenses:** $75.52M locked up due to over-purchasing and inefficient inventory levels.
2. **Operational Waste & Leakage:** $56.41M in absolute losses driven by Cold Chain Failures (34%), Damaged Stock (33.09%), and Expired Products (32.91%).

Technical Solution Deployed:

To eliminate these bottlenecks, an advanced Business Intelligence solution was built using a highly optimized **Star Schema Data Model** across 7 primary tables (3 Dimensions and 4 Fact Tables) in Power BI.

• **Data Engineering & Power Query:** Cleaned and transformed high-volume transactional data, enforcing referential integrity and strict 1:N cardinality relationships.

• **Advanced DAX Formulas:** Implemented complex context-aware metrics (such as SUMX for unit storage evaluation, DIVIDE with safety handling for Gross Margins, and lead-time-dependent Reorder Points).

• **Interactive Dashboards:** Developed four operational control towers tracking: Financial Performance, Waste Root-Causes, Stock Holding Status, and Supplier Lead-Time Efficiency.

Business Impact & Outcomes:

The framework transitions the business from reactive ordering to data-driven, demand-adjusted procurement. By implementing the suggested strategic actions—such as fixing cold chain infrastructure and moving to dynamic First-Expired, First-Out (FEFO) automated alerts—the organization can mitigate financial risk exposure, stop the $56M+ waste leakage, and successfully restore the business to positive profitability.